Fiscal Year:	FY 2005	**Task Last Updated:**	FY 09/12/2006
PI Name:	Lackner, James R. Ph.D.		
Project Title:	Adaptation in Artificial Gravity		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02454-9110	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-07
Start Date:	01/10/2003	**End Date:**	01/10/2007
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	DiZio, Paul (Brandeis University)		
Grant/Contract No.:	NAG9-01483		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We will determine whether arm, head, and object movement control can be adapted to rotational artificial gravity environments of 10 rpm. Earlier qualitative studies pointed to 5 rpm as the upper limit of adaptability. We have demonstrated, however, that reaching movements can be adapted to rotation at 10 rpm. Even without visual feedback, normal movement trajectories and endpoints are restored within 15-20 movements. Increases in background gravitoinertial force (GIF) level do not affect initial errors but affect rate and magnitude of adaptation. Decreases in GIF lead to smaller initial errors but adaptation proceeds at the same rate as in 1g. Adaptation of head movement control proceeds in two stages, rapid adaptation to Coriolis perturbations of the head as an inertial mass, slower adaptation to Coriolis perturbations of the vestibular system. Our present goals are to: 1) complete assessments of GIF influences on adaptation of arm movement control during rotation, 2) measure retention of arm movement adaptation and transfer of adaptation to different rotational velocities, 3) measure disruption and adaptation of head movement control and of 4) object transport as a function of rotational velocity. These studies would be conducted in our slow rotation room. The		

	first involves measuring reaching accuracy and adaptation as a function of the magnitude and direction of GIF for comparable Coriolis force perturbations. The second involves quantitative assessments of head movement disruption and adaptation as a function of SRR rotational velocity; the third involves assessments of object transport. The results will aid in specifying design criteria for artificial gravity environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study indicates that current attempts to rehabilitate patients with motor disorders by giving them training sessions on a robotic manipulandum generating a customized force field will not be successful, unless we can understand and circumvent the natural process of derecruiting the object component of a load when they release their grip on the manipulandum.
Task Progress:	This report covers the second year of this three year project. A previous report gave details of Year 1 progress, in which three major experiments were completed: 1. We evaluated adaptation of arm reaching movements to perturbations from a robotic manipulandum programmed to mimic Coriolis forces, when subjects had either a spatial goal or a an iso-effort goal. Successful adaptation was generated only with the spatial goal. This suggests that tasks involving force regulation will adapt slowly in a rotating environment but will also not be subject to aftereffects upon transition to a non-rotating environment. 2. We exposed subjects to a composite lateral force field, produced by a robotic manipulandum, consisting of a leftward velocity-dependent component plus a rightward constant force field. Subjects learned to move accurately after repeatedly reaching to a target in the composite force field, and then they displayed aftereffects opposite to its velocity-dependent component rather than opposite to its resultant force direction when both forces were removed. This study shows that internal models of dynamic forces are encoded in relative terms during motor learning, although the exact reference frame for relative coding remains to be determined. This will help determine whether generalized adaptation to an artificial gravity environment requires exposure to all possible combinations of artificial gravity level and Coriolis force or whether exposure to a limited subset of possible combinations will be sufficient. 3. We exposed subjects to composite contacting and non-contacting forces on the arm by having them reach while holding an object during constant velocity counterclockwise rotation. In a slow rotation room, a non-contacting, distributed, rightward Coriolis force acts on the arm and a rightward contact force is imposed by the object on the hand. Subjects adapted while holding the object and then did not overcompensate when initially reaching without it during rotation. We concluded that contacting and non-contacting forces are discriminated and learned separately. This indicates that astronauts in a rotating environment will need experience both moving their arm freely as well as wielding objects in order to do both without error in a rotating environment In the Year 2 of this project, we completed two new experiments which follow up the results of Item 3, described above. Experiment 3. Year 2 experiment I: We previously concluded that contacting and non-contacting forces are discriminated and learned separately. Here, we have tested an implication of this. When subjects adapt to contact forces from robotic manipulanda they show aftereffects when the force is turned off. We predicted that aftereffects would not occur if subjects removed their hand from the device thereby derecruiting the “object” component of the compensation before making free movements. Adaptation to the PHANToM was found to be device specific and represents adaptive tool use. These results have two important implications for countermeasures. First, subjects will have to be trained in a rotating artificial gravity environment to make unencumbered movements and to wield objects. Second, two forms or training can be done simultaneously, because the central nervous system separately adapts to the Coriolis forces on the body and the object/tool separately. Experiment 3. Year 2 experiment II: The second experiment attempts to understand the specificity/generalization of learning to wield an object/tool in a rotating environment. We started by having subjects learn to wield a single object in the rotating room and then assessing their performance with lighter and heavier objects. These results indicated that adaptation to one weight/tool transfers in a graded way to others similar weights. It may be possible to acquire a generalized adaptation to moving objects of a wide range of weights in a rotating artificial gravity environment by exposing subject with a few weights space throughout the operational range.
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